Restaurant Recommendation System

1.Intoduction : In today's digital world, users face **information overload** while choosing restaurants. This project aims to solve this problem by building a **Content-Based Restaurant Recommendation System** that suggests similar restaurants based on user input.

2.Objectives: The main objective of this project is:

- To recommend restaurants based on **user input (restaurant name)**
- To analyze **reviews and cuisines**
- To provide **similar restaurant suggestions**
- To sort recommendations based on **ratings**

3.Problem Statement: Users face confusion when choosing restaurants due to too many options and lack of proper guidance. Traditional searching takes time and may not give the best results. This system solves the problem by giving personalized recommendations based on restaurant data and reviews.

4.System Overview: This system uses a **Content-Based Filtering approach**, where:

- Input: Restaurant name + user location
- Process: Analyze reviews using NLP
- Output: List of similar restaurants

5.Technologies Used :

- **Python** – for backend logic
- **Flask** – for building web application
- **Pandas** – for data processing and cleaning
- **Scikit-learn** – for machine learning model
- **HTML & CSS** – for designing frontend

6.Dataset:

The project uses the **Zomato dataset**, which contains:

- Restaurant name
- Ratings
- Cuisines

- Reviews
- Cost for two people
- Location

This data helps the system understand restaurant features and user preferences

7.Data Processing:

- Removed missing (null) values
- Removed duplicate records
- Cleaned rating values (converted to float)
- Converted cost column into numeric format
- Converted text to lowercase
- Removed punctuation from reviews

These steps improve the accuracy of the model

8.Model Used:

- **TF-IDF Vectorization**

  Converts text data (reviews) into numerical form
- **Cosine Similarity**

  Measures similarity between restaurants based on reviews

This combination helps in finding restaurants with similar features.

9.Working of System

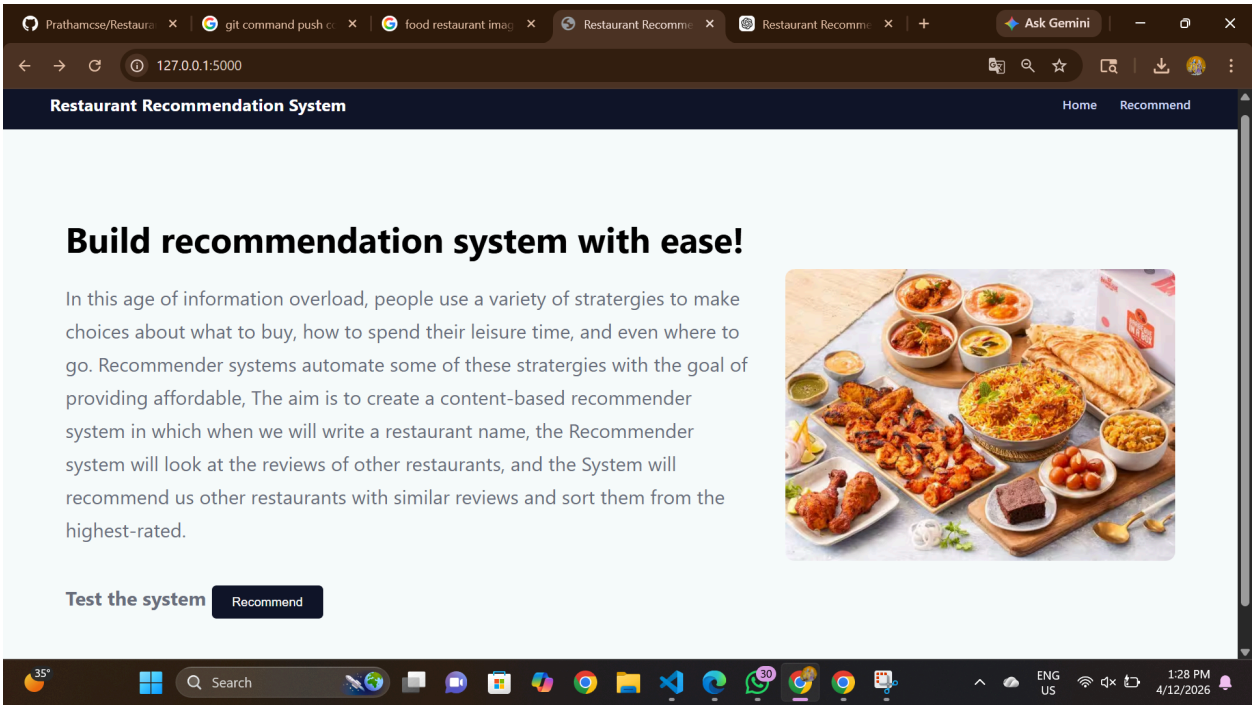
1. User enters a restaurant name
2. System searches the dataset
3. Finds similar restaurants using cosine similarity
4. Filters results (optionally by location)
5. Displays top recommended restaurants

10.Web Application

The project includes 3 main pages:

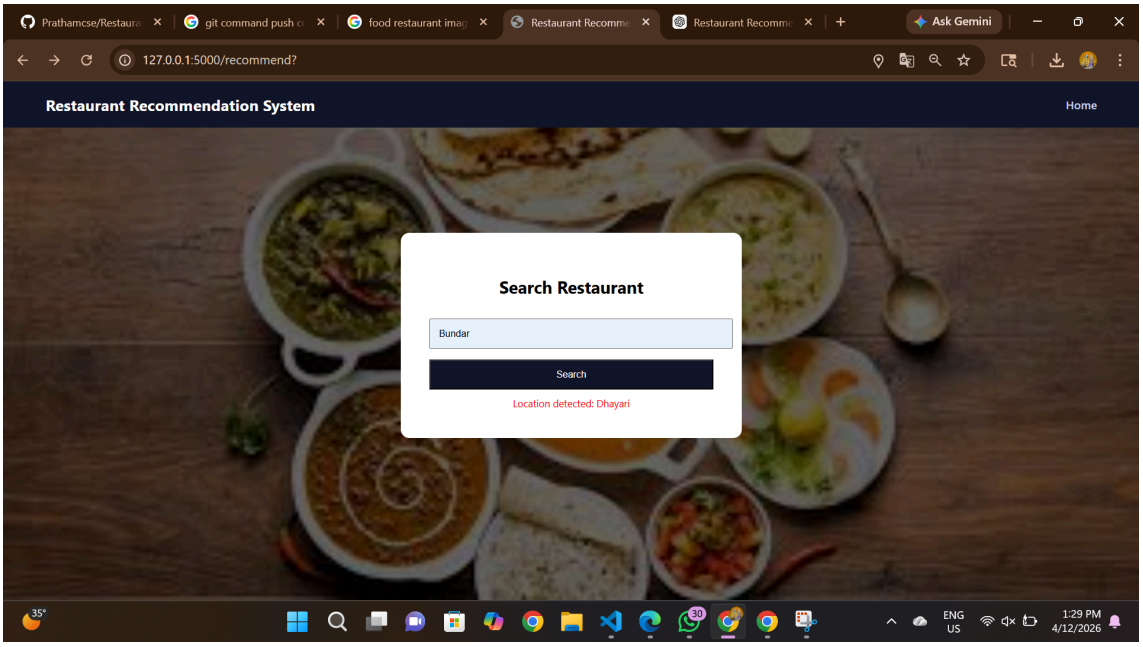
- **Home Page**

Shows project overview and navigation



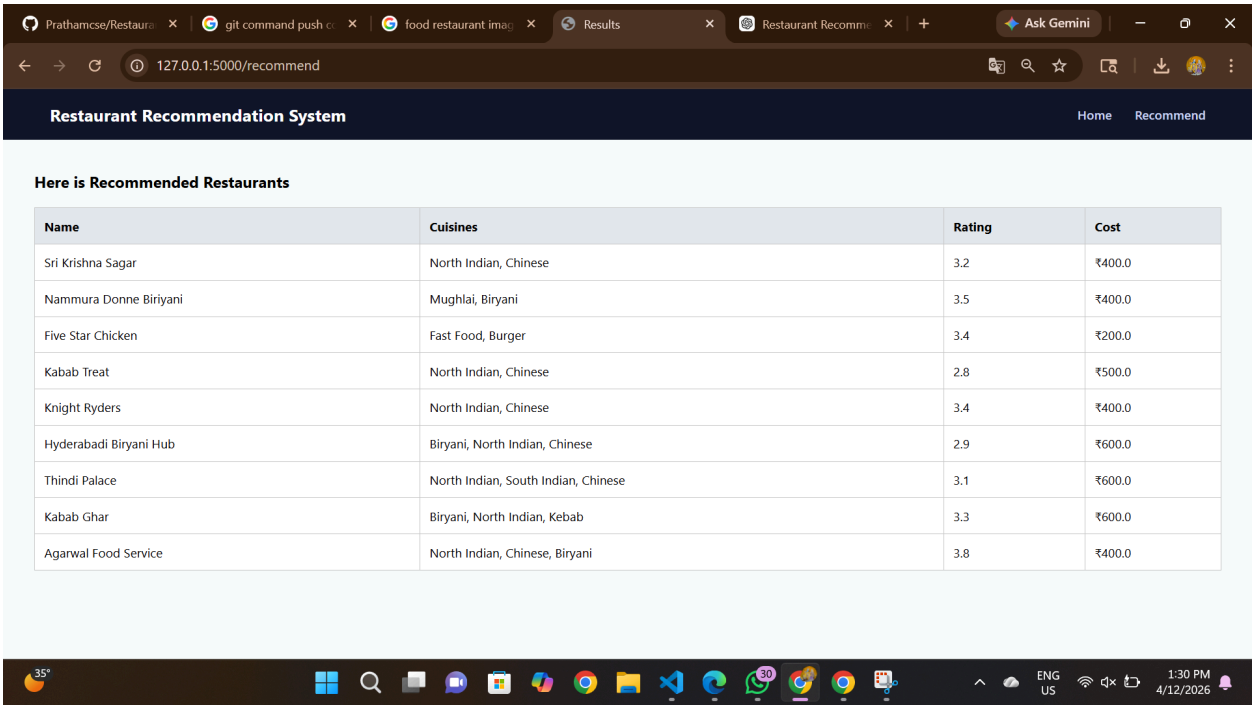
- **Recommendation Page**

User enters restaurant name



- **Result Page**

Displays recommended restaurants in table format



Name	Cuisines	Rating	Cost
Sri Krishna Sagar	North Indian, Chinese	3.2	₹400.0
Nammura Donne Biriyani	Mughlai, Biryani	3.5	₹400.0
Five Star Chicken	Fast Food, Burger	3.4	₹200.0
Kabab Treat	North Indian, Chinese	2.8	₹500.0
Knight Ryders	North Indian, Chinese	3.4	₹400.0
Hyderabadi Biryani Hub	Biryani, North Indian, Chinese	2.9	₹600.0
Thindi Palace	North Indian, South Indian, Chinese	3.1	₹600.0
Kabab Ghar	Biryani, North Indian, Kebab	3.3	₹600.0
Agarwal Food Service	North Indian, Chinese, Biryani	3.8	₹400.0

11. Output

The system shows:

- Restaurant Name
- Rating
- Cuisines
- Cost for two people

This helps users compare and choose easily

12.Features:

- Fast recommendation system
- Uses real-world dataset
- Shows multiple restaurant details
- Easy navigation between pages